GATE ALL BRANCHES



Probability and Statistics

-ENGINEERING MATHEMATICS

Lecture No.- 07



By- Rahul sir

Recap of Previous Lecture



Topic Classification of events

Topic Conditional probability

Topic Bayes theorem

Topics to be Covered



Topic Classification of events

Topic Problems based on Conditional probability

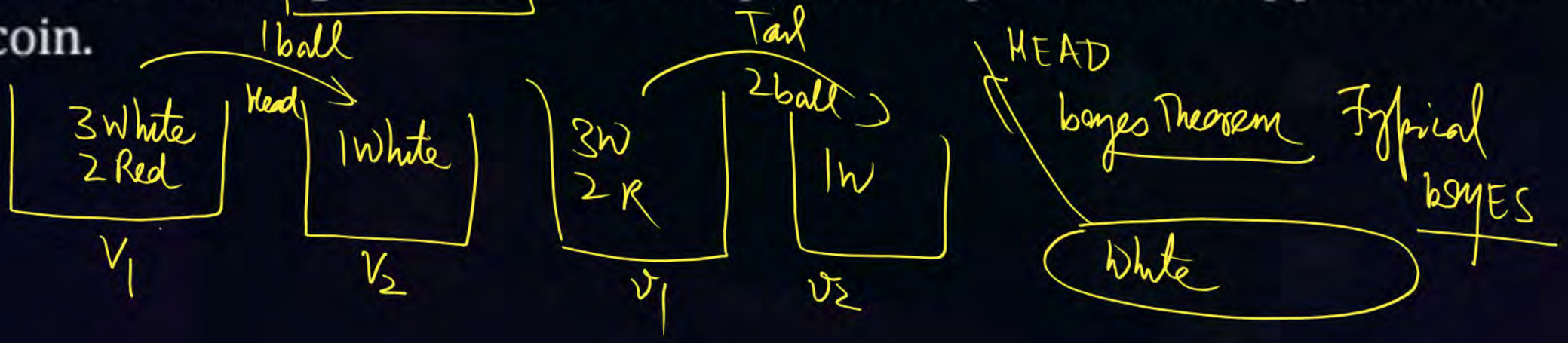
Topic Problems based on Bayes theorem

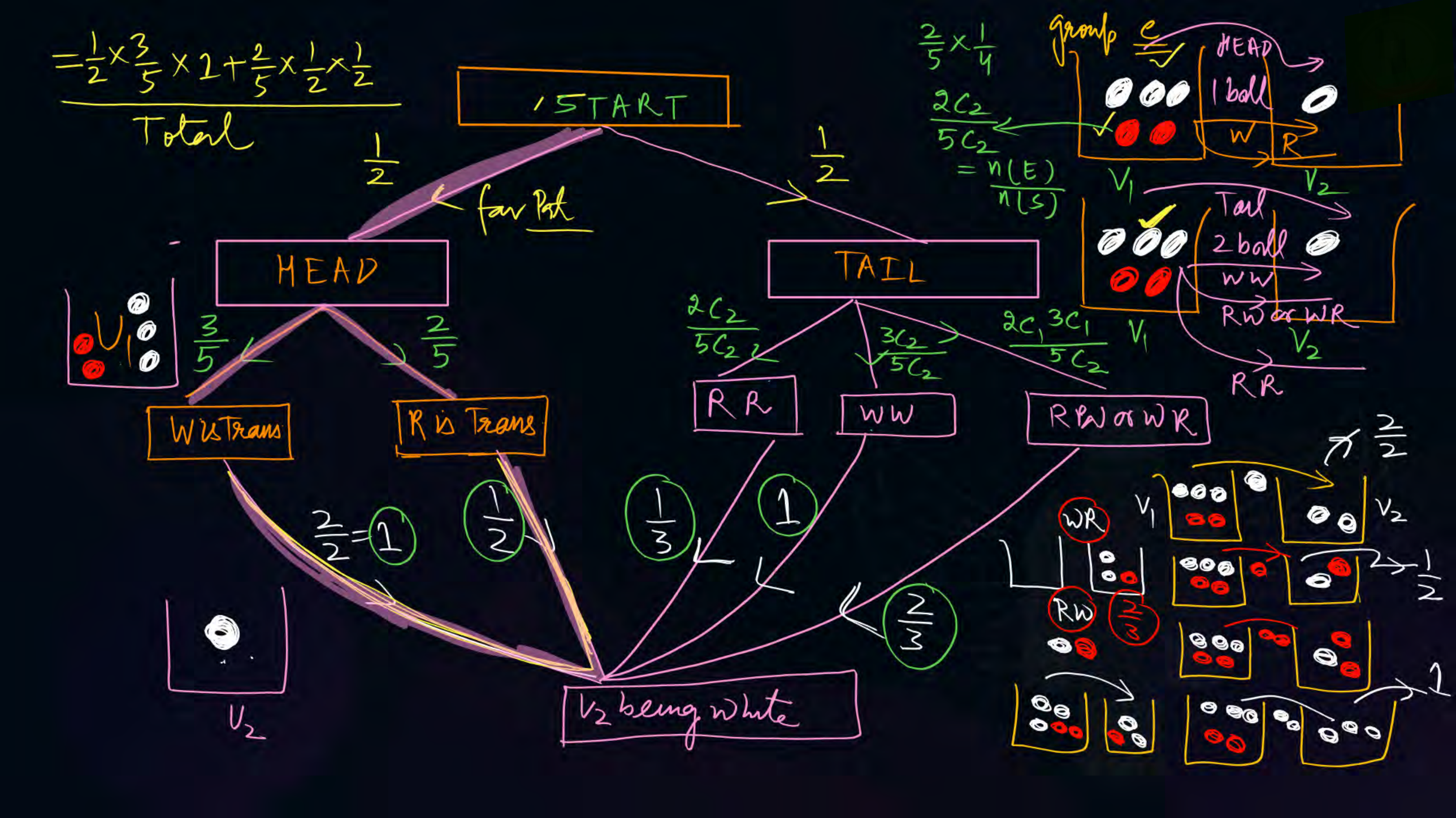


Topic : Probability and Statistics

$\frac{12}{23}$

#Q. Let v_1 and v_2 be two urns box such that v_1 contains 3 White and 2 Red balls and v_2 contains only 1 White ball. A fair coin is tossed. If head appears then 1 ball is drawn at random from V_1 and put Into v_2. However If tail appears then 2 balls are drawn at random from v_1 and put into v_2. Now one ball is drawn at random from v_2 given that the drawn ball from v_2 is white then the probability that head appeared on the coin.




$= \frac{\frac{1}{2} \times \frac{3}{5} \times 1 + \frac{2}{5} \times \frac{1}{2} \times \frac{1}{2}}{\text{Total}}$
START
$\frac{1}{2}$
$\frac{1}{2}$
fav Path
HEAD
TAIL
U_1
$\frac{3}{5}$
$\frac{2}{5}$
W is Trans
R is Trans
$\frac{2}{2} = 1$
$\frac{1}{2}$
U_2
$\frac{2C_2}{5C_2}$
$\frac{3C_2}{5C_2}$
$\frac{2C_1 3C_1}{5C_2}$
RR
WW
RW or WR
$\frac{1}{3}$
1
$\frac{2}{3}$
V_2 being white
$\frac{2}{5} \times \frac{1}{4}$
$\frac{2C_2}{5C_2} = \frac{n(E)}{n(S)}$
group C
HEAD
1 ball
W
R
V_1
V_2
Tail
2 ball
WW
RW or WR
RR
WR
RW
$\frac{2}{2}$
$\frac{1}{2}$
1

$$= \frac{\frac{1}{2} \times \frac{3}{5} \times 1 + \frac{1}{2} \times \frac{2}{5} \times \frac{1}{2}}{\frac{1}{2} \times \frac{3}{5} \times 1 + \frac{1}{2} \times \frac{2}{5} \times \frac{1}{2} + \frac{1}{2} \times \frac{{}^{2}C_{2}}{{}^{5}C_{2}} \times \frac{1}{3} + \frac{1}{2} \times \frac{{}^{3}C_{2}}{{}^{5}C_{2}} \times 1 + \frac{1}{2} \times \frac{{}^{3}C_{1} \times {}^{2}C_{1}}{{}^{5}C_{2}} \times \frac{2}{3}}$$

$$P\left(\frac{\text{HEAP app. com}}{V_2 \text{ being White}}\right) = \frac{12}{23}$$ Ans



Topic : Probability and Statistics

#Q. In a housing society, half of the families have single child per family, while the remaining half have two children per family. The probability that a child picked at random, has sibling is_____.

(2 child)

N families

$\frac{N}{2} \times 1$

$\frac{N}{2} \times 2$ favourable

$$P(E) = \frac{n(E)}{n(S)} = \frac{\frac{N}{2} \times 2}{\frac{N}{2} \times 2 + \frac{N}{2} \times 1} = \frac{2}{3}$$

$= 5$
$= 10$

$$\frac{10}{10+5} = \frac{2}{3}$$



Topic : Probability and Statistics

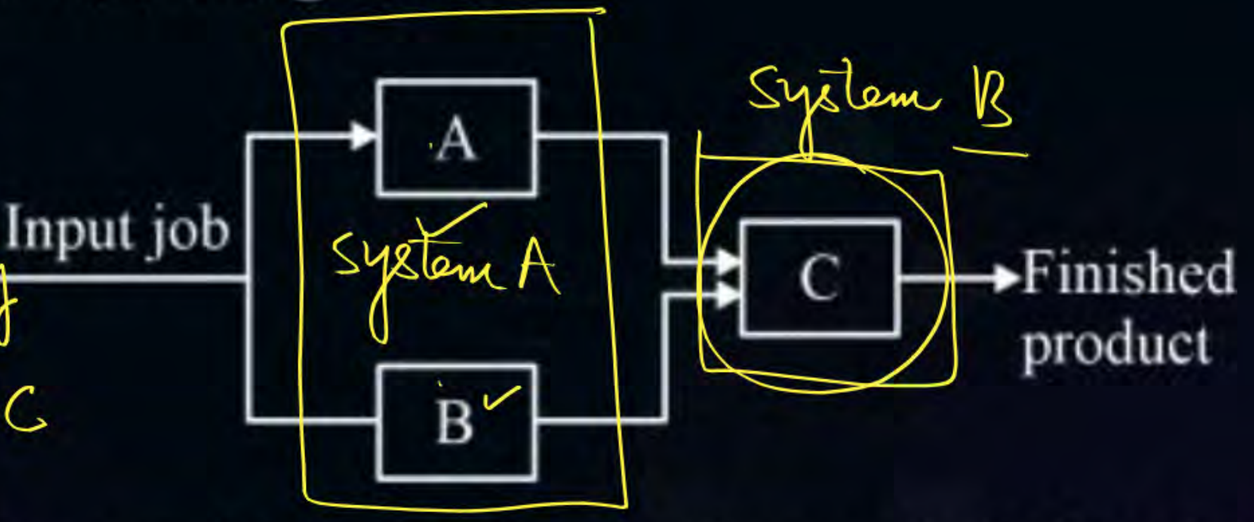
A, B, C – Independent

#Q. The figure shown the schematic at a production process with machines A, B and C. An input job needs to be pre-processed either by A or B by before it is fed to C, form which the final finished product comes out. The probabilities of failure the machines are given as:

$P_A = 0.15$, $P_B = 0.05$ & $P_C = 0.1$

P(A ∪ B ∩ C) = successfully

[A or B] and C A or B and C (A ∪ B ∩ C)



Assuming independence of failures of the machines, the probability that a given job is successfully processed (up to the third decimal place) is

$P(A) = 0.15 = P(\bar{A}) = 1 - 0.15 = 0.85$

$P(B) = 0.05 \quad P(\bar{B}) = 1 - 0.05 = 0.95$

$P(C) = 0.01 \quad P(\bar{C}) = 1 - 0.1 = 0.90$

(A), (B) $\longrightarrow$ C

$(A \cup B) \cap C$

$P(\bar{A} \cup \bar{B} \cap \bar{C})$

$P(X \cup Y \cap Z)$

$\bar{A} = P(X)$

$\bar{B} = P(Y)$

$\bar{C} = P(Z)$

$$P(X \cup Y \cap Z) = P(X \cup Y)\, P(Z)$$

$$= [P(X) + P(Y) - P(X \cap Y)]\, P(Z)$$

$$= [(P(X) + P(Y) - P(X)P(Y)]\, P(Z)$$

$$= [0.85 + 0.95 - 0.85 \times 0.95][0.90 \Rightarrow \boxed{0.982}$$



Topic : Probability and Statistics

$P(G) = \frac{4}{5}$ $P(R) = \frac{1}{5}$

Read The Problem

#Q. A signal which can be green or red with probability 4/5 and 1/5 respectively received by station A and then transmitted to station B. The probability of each station receiving the signal correctly is 3/4. If the signal received at station B is green the probability that the original signal was green.

original — A ⟶ B

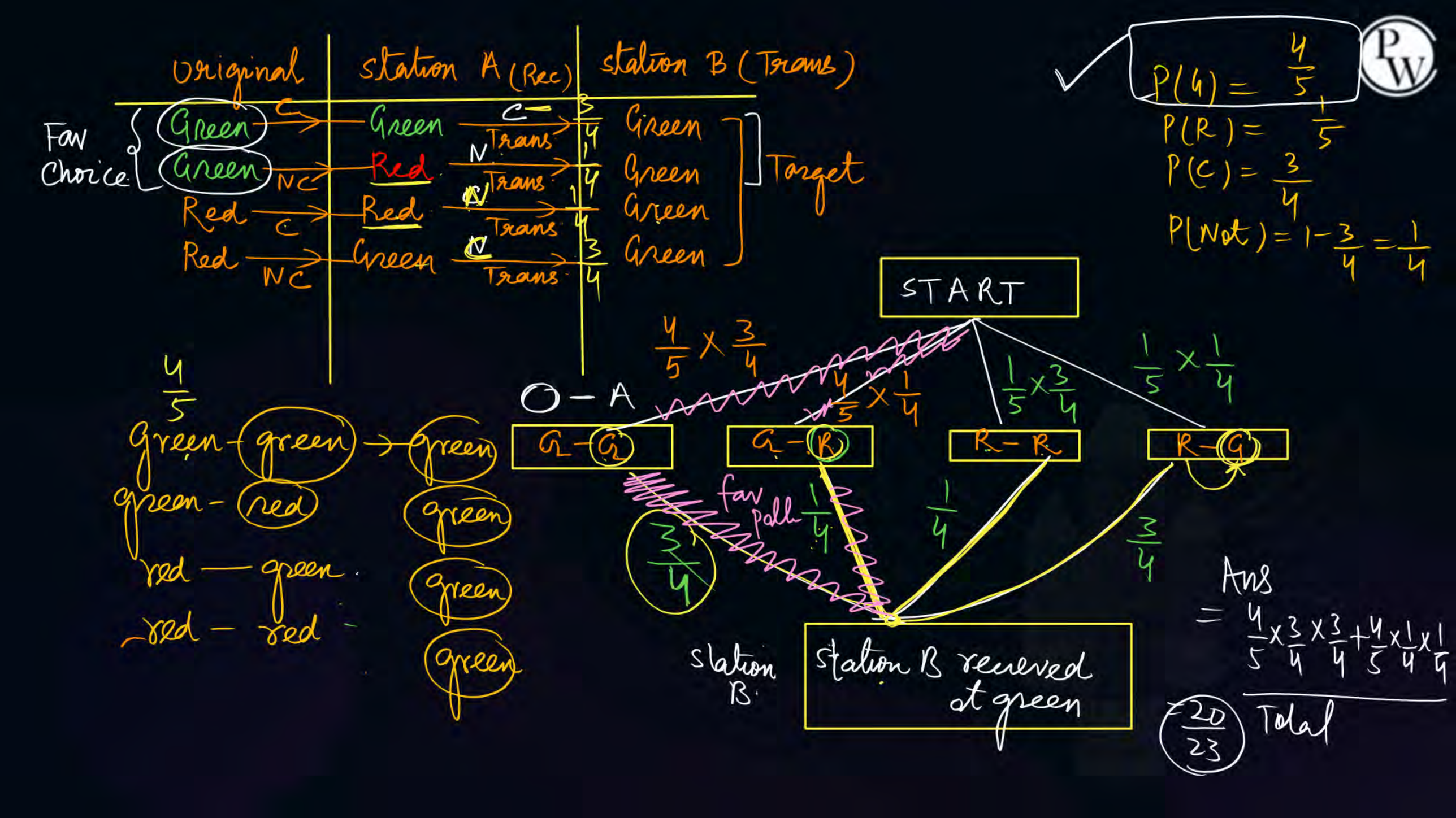
correct $\frac{3}{4}$

A 3/5

B 6/7

C 20/23

D 9/20


Original
station A (Rec)
station B (Trans)
Fav Choice
Green
Green
Red
Red
Red
C
NC
C
NC
C
Trans
NC
Trans
C
Trans
NC
Trans
Green
Red
Red
Green
Green
Green
Green
Green
Target
3/4
1/4
1/4
3/4
P(G) = 4/5
P(R) = 1/5
P(C) = 3/4
P(Not) = 1 - 3/4 = 1/4
4/5
green - green → green
green - red
red — green
red - red
green
green
green
green
O - A
START
4/5 x 3/4
4/5 x 1/4
1/5 x 3/4
1/5 x 1/4
G - G
G - R
R - R
R - G
3/4
fav path
1/4
1/4
3/4
station B
station B received at green
Ans = 4/5 x 3/4 x 3/4 + 4/5 x 1/4 x 1/4 / Total
20/23



Topic : Probability and Statistics

#Q. If P (X)= ¼, P(Y) = 1/3 and P(X ∩ Y) = 1/12,then value of P (Y/X) is-

$P(X) = \frac{1}{4}$ $P(X \cap Y) = \frac{1}{12}$

$P(Y) = \frac{1}{3}$ $P\left(\frac{Y}{X}\right) = ?$

$\frac{P(X \cap Y)}{P(X)} = P\left(\frac{Y}{X}\right)$

$\frac{\frac{1}{12}}{\frac{1}{4}} = P\left(\frac{Y}{X}\right)$

$P\left(\frac{Y}{X}\right) = \frac{1}{3}$

A 1/4

B 4/25

C 1/3

D 29/50



Topic : Probability and Statistics

P(R)(1-P(Ramesh) + P(ramesh)(1-ram)

#Q. Ram and Ramesh appeared in an interview for two vacancies in the same department. The probability of Ram's selection is 1/6 and that of Ramesh is 1/8. What is the probability that only one of them will be selected?

$P(Ram) = \frac{1}{6}$

$P(ramesh) = \frac{1}{8}$

Exactly one Selected

Ram Ramesh

$= P(Ram)\,P(\overline{Ramesh}) + P(ramesh)\,P(\overline{Ram})$

$= \frac{1}{6} \times \left(1-\frac{1}{8}\right) + \frac{1}{8}\left(1-\frac{1}{6}\right)$

$= \frac{1}{6} \times \frac{7}{8} + \frac{1}{8} \times \frac{5}{6}$

$= \frac{12}{48} = \left(\frac{1}{4}\right)$

A 47/18

B 1/4 ✓

C 13/48

D 35/48



Topic : Probability and Statistics

H.W

#Q. The probabilities that a student passes in Mathematics, Physics and Chemistry are m, p and c respectively. Of these subjects, the student has 75% chance of passing in at least one, a 50% chance of passing in at least two and a 40% chance of passing in exactly two. Following relations are drawn in m, p, c:

I. $p + m + c = 27/20$

II. $p + m + c = 13/20$

III. $(p) \times (m) \times (c) = 1/10$

$p+m+c = \frac{27}{20}$

$pmc = \frac{1}{10}$

A. Only relation I is true

B. Only relation II is true

C. Relations II and III are true

D. Relations I and III are true



Topic : Probability and Statistics

= H·W GATE

#Q. P and Q are considering to apply for a job. The probability that P applies for the job is 1/4, the probability that P applies for the job given that Q applies for the job is 1/2, and the probability that Q applies for the job given that P applies for the job is 1/3. Then the probability that P does not apply for the job given that Q does not apply for this job is-

Conditional prob.

A ✓ 4/5

B 5/6

C 7/8

D 11/12

THANK - YOU